

Kayoon Kim

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RESEARCH INTEREST

Human-AI Interaction, Decision Making, Computational Social Science

EDUCATION

University of Tübingen, Tübingen, Germany 2023.10-2026.03

M.S., Quantitative Data Science (*Concentration*: Machine Learning), GPA 1.81/1.0

Thesis: Sim-DSE: Mediating Multi-User Orchestration in Confined Shared Spaces through Simulation-Augmented Design Space Exploration (Advisor: Jan Henry Belz, Prof. Holger Brandt)

Chung-Ang University, Seoul, South Korea 2017.03-2023.02

B.A. B.E., Sociology & Computer Science, GPA 4.13/4.5 & 3.96/4.5

Thesis: Claiming for Rights: The Changing Landscape of Issues and Claims in the National Petition of South Korea, 2017-2020 (Advisor: Prof. Chan S. Suh)

RESEARCH EXPERIENCE

Thesis Student, Porsche AG Human-Centered AI Research (Advisor: Jan Henry Belz) 2025

- Identifying in-vehicle interaction design space for multi-occupant autonomous vehicles to establish design guidelines for the field; conducting a simulation study and an online survey
- Engineering a context-aware multi-agent architecture that enables adaptive vehicle configuration and human-centered interaction design, with an expectation for a user-study

Research Intern, Bosch Center for Artificial Intelligence (Advisor: Lukas Grossberger) 2024

- Developed and tested Preferential Bayesian Optimization algorithms to efficiently optimize machinery settings for expert operators, modeling human preferences in decision-making processes
- Conducted synthetic bias experiment, identified cognitive biases affecting decision-making and developed mitigation strategies for future human-in-the-loop experimental design

Research Assistant, Methods Center (Advisor: Prof. Augustin Kelava) 2023-2024

- Developed the model to predict student dropouts in STEM classes using multivariate time-series analysis (i.e., Kalman Filter) to strengthen the educational environment

Research Assistant, [Data Science Lab](#) (Advisor: Prof. Yoonsik Cho) 2021

- Explored recommendation systems by implementing and evaluating algorithms to enhance human decision-making

Research Assistant, [HIKE Lab](#) (Advisor: Prof. Haklae Kim) 2020-2021

- [COVID-19: Our Memory](#): Identified pandemic-driven social inequalities through statistical correlation analysis, text mining, and archival research; recognized with a Community Award in the collaborative hackathon organized by the Ministry of Interior and Safety of Korea, among 320 participants

- Assessed policy efficacy (i.e., screening facilities, hospital bed capacity), enhanced data visualization accessibility for the general public, and developed recommendations for public health data infrastructure improvements as a team; presented the work on International Open Data Day 2021

- As a first case of a non-governmental digital archive, both projects were donated to the National Library of Korea

Research Assistant, [Da Move Lab](#) (Advisor: Prof. Chan S. Suh) 2020-2023

- Constructed and analyzed 3-year dataset of South Korea's online petitions revealing embedded biases through text mining and NLP techniques
- Extended dataset analysis to entire system's duration; applied fine-tuned language models for sentiment analysis and regression modeling to identify structural limitations in civic engagement platforms

PAPERS	[2] Sim-DSE: Mediating Multi-User Orchestration in Confined Shared Spaces through Simulation-Augmented Design Space Exploration Jan Henry Belz*, Kayoon Kim * <i>In progress, IMWUT 2026</i> *Equal Contribution	
	[1] Mobilizing grievances in the internet age: The case of national online petitioning in South Korea, 2017–2022 Kayoon Kim , Chan S. Suh PLOS ONE 2024	
POSTERS	[1] Claiming for Rights: The Changing Landscape of Issues and Claims in the National Petition of South Korea, 2017-2020 Kayoon Kim International Postgraduate and Academic Conference 2021	
WORK EXPERIENCE	Data Quality Manager Intern , Kakao Style, South Korea - Analyzed model prediction data and developed dashboards to enhance model accuracy and user experience - Designed and developed an automatic labeling model that automated labeling processes, reducing manual effort by 50%	2023
HONORS & GRANTS	DAAD STIBET Scholarship , German Academic Exchange Service (DAAD)	2025
	Research Grant , Porsche AG	2025-2026
	Research Grant , National Research Foundation of Korea	2024
	Merit-and-need-based Scholarship , Seoul Scholarship Foundation	2022
	Merit-and-need-based Scholarship , Lotte Scholarship Foundation	2021
	Community Award , Ministry of the Interior and Safety, South Korea	2020
SERVICE	National Scholarship , Korea Student Aid Foundation	2017-2021
	Communication Chair Volunteer, Tübingen Women in Machine Learning	2025
	Content Creator Volunteer, Soapbox Science Tübingen	2025
	Guest Speaker, Chung-Ang University Delivered a single session lecture on the overview of computational social science and data science for an Introduction to Sociology course to engage freshmen	2022
	Academic Advisor, COSADAMA Designed instructional sessions on Git, GitHub, and Python programming for non-STEM major students; mentored participants throughout the learning process	2020-2021
	Class Representative, Chung-Ang University	2018
SKILLS	Language English (Full professional proficiency), Korean (Native) Programming and Development Python, R, C, SPSS, MySQL, Pytorch, Git, Linux	